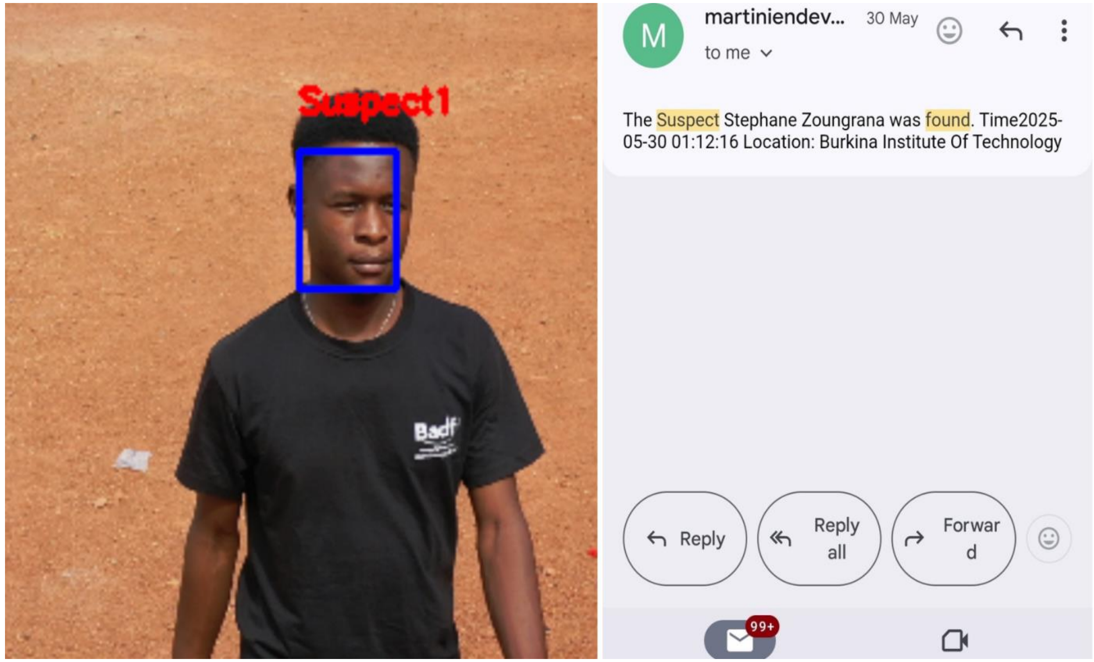
Bit Students Bachelor Thesis Topics Presentation

16/10/2025

Project Presentation

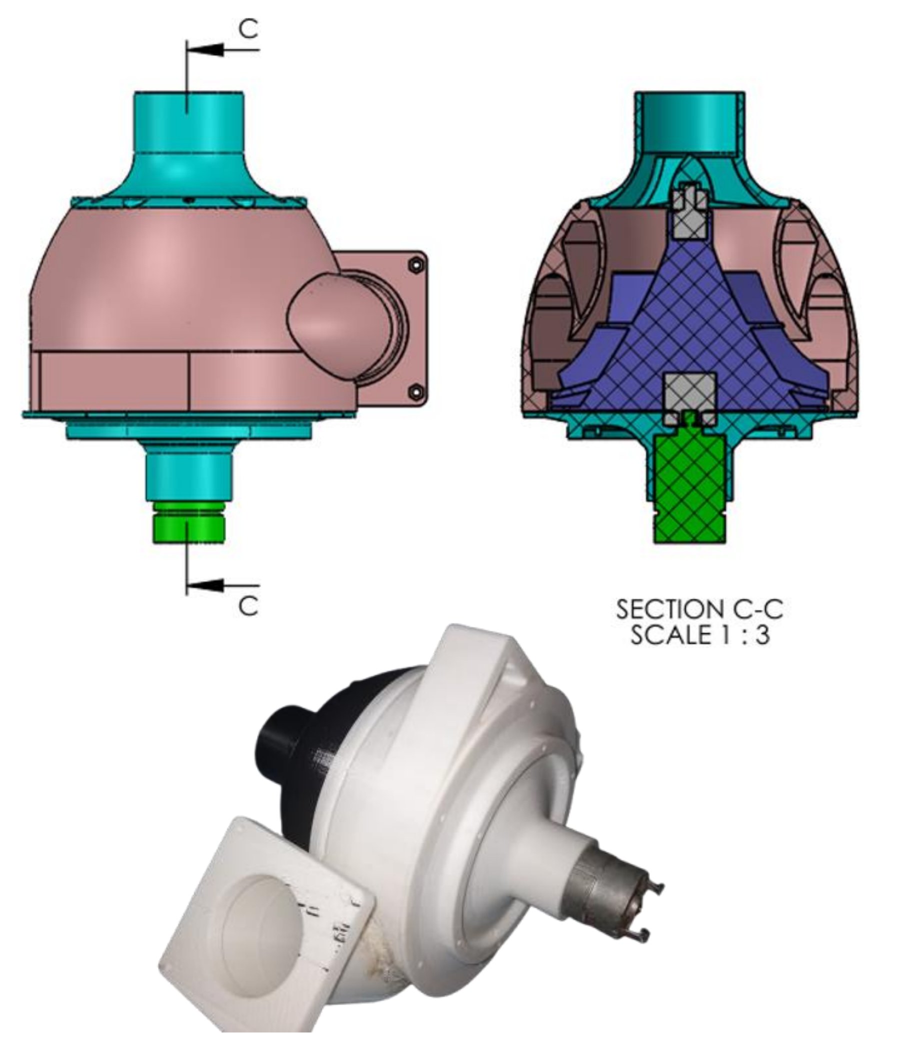
- **Project name:** a suspect Facial Recognition System for Closed Circuit Television Integrations with instant alerts
- **Problem Statement:** Burkina Faso faces ongoing terrorist threats. Existing Smart Burkina CCTV systems are ineffective due to weak real-time processing, incomplete suspect databases, and privacy issues, limiting their ability to identify and track suspects efficiently.
- **Innovative Solution:** Use of a drone to simulate CCTV perspectives. Encrypted alert system delivering GPS-based suspect location and timestamps.
- **Expected Outcome:** Enhanced national security, contributing to Burkina Faso’s counter-terrorism strategy. Scalable and ethical surveillance system respecting privacy.



SALOWE CEDRIC MARTINIEN ADOUABOU

Project Presentation

Topic : *Designing an aeration system suitable for small-scale mines*



❑ Problem Statement

- **Poor air quality**
- **High risk of asphyxiation and lung disease**
- **Excessive heat and physical discomfort**
- **The high cost of existing ventilation systems**

❑ Innovative Solution

The project proposes **a solar-powered air pump** that operates with an electric motor to supply fresh air into mining shafts. The system is designed to be low-cost, energy-efficient, and easy to maintain, making it ideal for remote small-scale mining sites.

❑ Expected Outcome

The system will ensure a steady flow of clean air, improving safety, reducing health hazards, and promoting sustainable mining through the use of renewable solar energy.

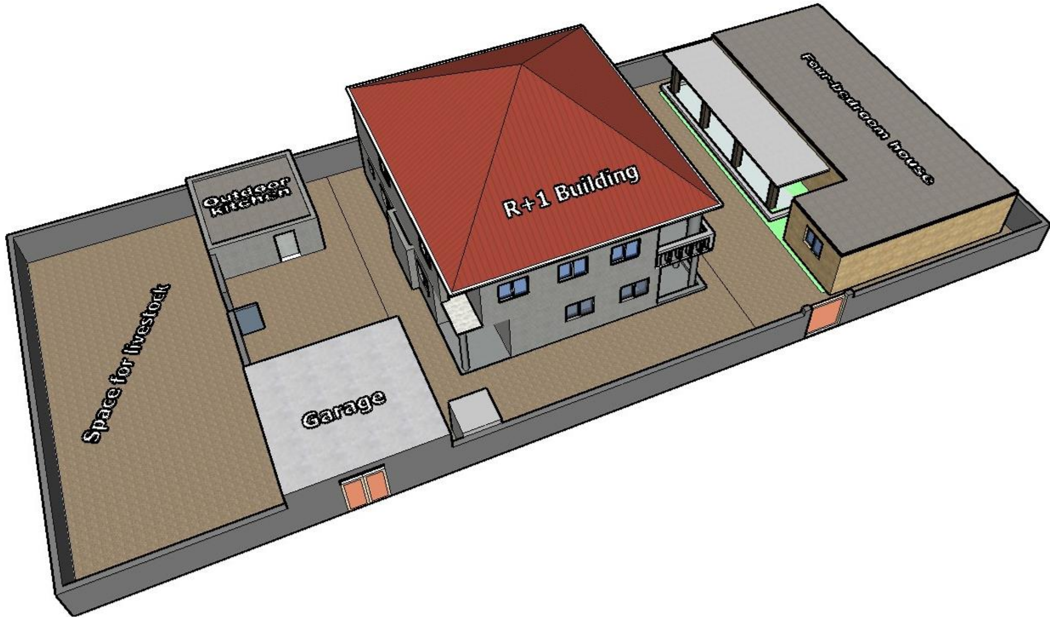
Project Presentation

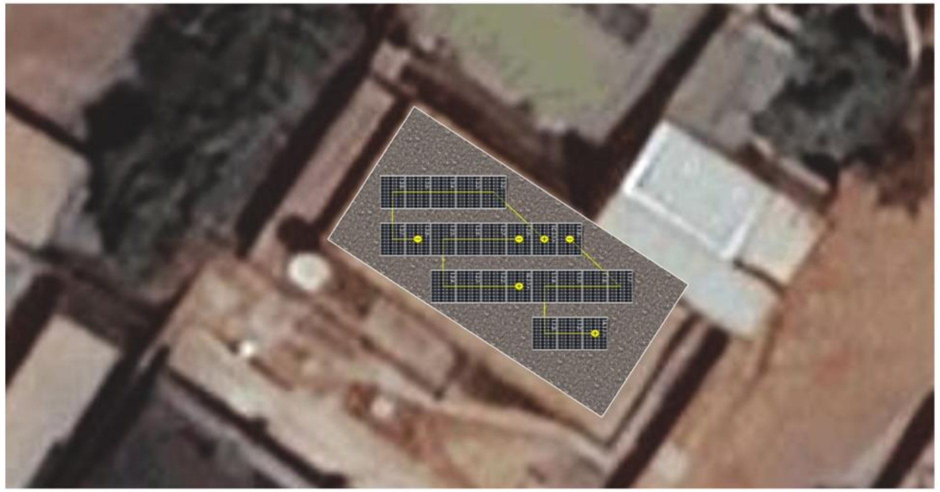
- **Project name :** Sizing and Design of a Solar Photovoltaic System for a Two-Storey Residential Building in Ouagadougou
- **Problem Statement :** Frequent power cuts and limited access to reliable electricity from the national grid (SONABEL) hinder comfort and energy security in residential buildings
- **Innovative Solution :** Design and implementation of an autonomous solar photovoltaic system to ensure continuous energy supply
- **Expected Outcome :**

✓ Reliable and sustainable power supply

✓ Reduction of electricity costs

✓ Possibility to power extra loads (water pump, guest rooms)





Project Presentation

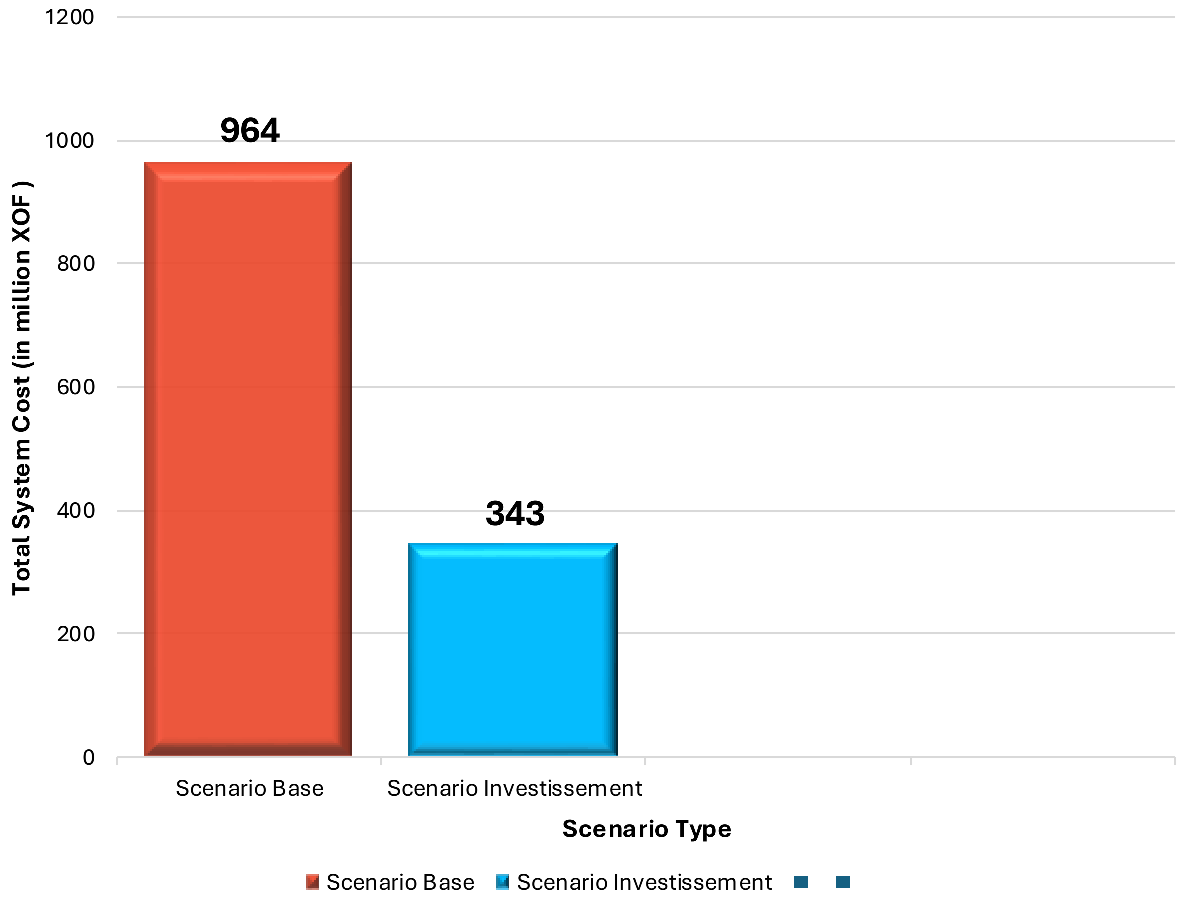
- **Analysis of electricity System Of Burkina Faso**

- How can we optimize the operation of the national electricity system in a context of growing demand energy and transition to renewable energies, while ensuring its reliability, accessibility, and sustainability**?**

- **Integration of 200 MWp solar capacity** across Ouagadougou, Bobo-Dioulasso and Pa to boost local renewable production; **Deployment of 50 MW battery storage** to stabilize the grid and supply power during evening demand peaks.

- **Total system cost reduction by about 60 %** (from 964 M FCFA to 344 M FCFA per year).

- **Energy autonomy raised to ≈ 70 %** while reducing electricity imports to less than 40 %.



Project Presentation

- ## Project name

Design and implementation of a mobile application for university events management.

- ## Problem Statement

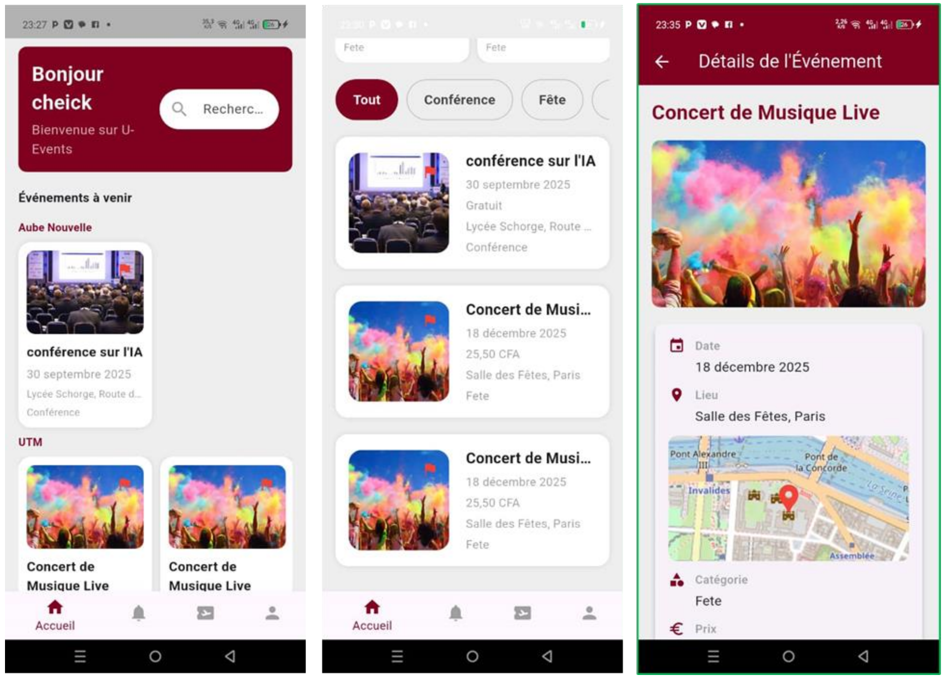
How could we design an intuitive and interactive mobile app that would enable the organization, promotional and involvement in university events and with the promised user experience?

- ## Innovative Solution

The university event management app helps universities organize and manage events efficiently. It lets students discover, register for, and get updates on events, while organizers can plan and handle registrations. It also connects universities by sharing events across institutions.

- ## Expected Outcome

facilitate communication, planning and participation in university events. Centralize information, reduce costs and errors, promote collaboration between universities and make activity management more modern, faster and efficient.



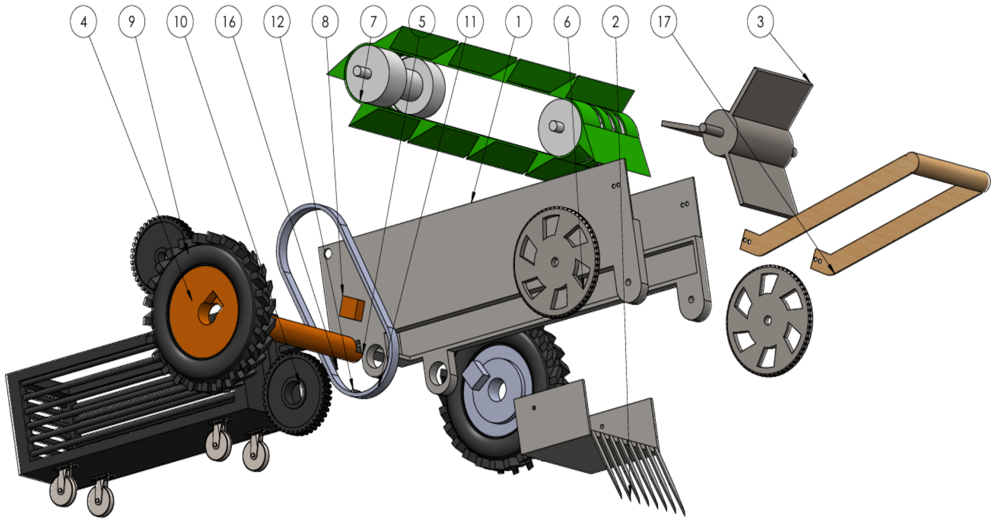
Project Presentation

- **Project name :** Multifunctional harvesting for Onion, Carrot, Lettuce.

- **Problem Statement:** Lack of mechanisation, Difficult access to imported machinery, Inadequacy of machinery and Post-harvest losses.

- **Innovative Solution:** An affordable, locally made harvester to efficiently collect onions, carrots, and lettuce

- **Expected Outcome:** Reduce labor costs and harvesting time for smallholder farmers and reduce crop losses and improve post-harvest efficiency.



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1	Chassis soutenance	1
2	Socs	1
3	Tambour ramasseur	1
4	Assemblage3 roue	2
5	Arbre	1
6	Roue avant	2
7	Assemblage convoyeur	1
8	clavette 2	2
9	poulie 1	1
10	poulie 2	1
11	Courroie1-3^VUE ECLATé	1
12	Pièce5^VUE ECLATé	1
16	Assemblage tremis	1

Project Presentation

- **Project name:** "Design of a smart meter with current leakage detection and remote control options".
- **Problem statement:** Inefficient energy management and current leakage due to obsolete installations.
- **Innovative solution:** My meter allows users to remotely control their appliances, detect current leakages and monitor consumption through a mobile app. The system alerts users in case of anomalies, displays energy consumption costs in real time.
- **Expected outcome:** Empower every household to save energy, reduce bills, and ensure electrical safety all through a single connected device.